

Table 1. 2022 independent-test performance

No	Input	Random Forest MAE	Random Forest RMSE	Random Forest PCC	XGBoost MAE	XGBoost RMSE	XGBoost PCC	LightGBM MAE	LightGBM RMSE	LightGBM PCC	CatBoost MAE	CatBoost RMSE	CatBoost PCC
1	Comb1	69.74	99.24	0.697	67.97	98.98	0.698	68.60	97.72	0.707	65.32	94.50	0.732
2	Comb1_land	63.88	91.44	0.756	65.51	93.62	0.742	63.40	92.02	0.744	59.03	87.98	0.780
3	Comb2	66.22	94.55	0.741	66.36	96.77	0.727	65.76	94.87	0.739	62.03	95.99	0.726
4	Comb2_land	59.32	90.29	0.760	61.59	91.34	0.764	57.10	88.03	0.779	55.86	87.88	0.776
5	Comb3	63.91	93.03	0.735	68.48	100.64	0.701	71.96	101.53	0.704	67.63	97.11	0.713
6	Comb3_land	58.51	88.49	0.756	65.55	99.02	0.711	65.55	97.45	0.715	61.24	93.91	0.729
7	CCS_land	58.99	93.57	0.729	66.12	98.35	0.706	63.62	92.40	0.740	61.03	93.40	0.729
8	CDR_land	59.96	95.66	0.711	71.80	111.56	0.625	68.52	105.46	0.663	60.94	98.16	0.700
9	CHIRPS_land	56.67	88.85	0.769	63.46	94.09	0.745	63.94	90.92	0.768	54.96	85.92	0.785
10	ERA5_land	56.18	88.21	0.775	66.45	96.41	0.718	60.55	88.68	0.774	58.51	88.49	0.767
11	GSMAP_GAUGE_land	60.62	93.94	0.763	61.47	93.67	0.779	67.62	100.40	0.726	65.36	97.91	0.737
12	GSMAP_MVK_land	65.89	106.73	0.652	72.82	110.77	0.627	73.82	112.97	0.616	69.98	106.24	0.660
13	IMERG_land	58.53	88.68	0.756	62.64	90.94	0.750	63.95	93.55	0.738	60.32	88.88	0.760
14	PDIR_land	61.52	95.91	0.717	66.17	99.32	0.703	69.20	103.59	0.684	63.48	94.22	0.730
15	PERSIANN_land	67.58	119.60	0.546	73.45	118.36	0.563	67.12	114.04	0.603	68.33	109.80	0.628